MIL-A-21672(NOrd)

21 November 1958
SUPERSEDING
NAVORD OS 2677A
28 May 1945

MILITARY SPECIFICATION

ARMING CELL MARK 2 MODS 1 AND 2

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the Arming Cell Mark 2 Mod 1 and Mod 2. The arming cell is a delay-arming device for use on clock starters and for limited use on extenders in place of soluble washers. Electrolytic action between a zinc disk and a cup of silver chloride, arising from sea water immersion, causes the disk to deteriorate until it breaks and releases the piston of the clock starter or extender.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Boxes, Fiber

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-6872	Soldering Processes, General Specification for.
MIL-C-10464	Cans, Hermetic Sealing, Metal Light Gage Tear-Strip Type.

FSC 1350

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

BUREAU OF ORDNANCE

LD 139380	Arming Cell Mark 2 Mod 1
LD 139381	Arming Cell Mark 2 Mod 2

And all documents listed in the above Lists of Drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General requirements.

3.1.1 The Arming Cell Mark 2 Mods 1 and 2 shall be manufactured in accordance with applicable Lists of Drawings and all documents listed thereon.

Arming Cell Mark 2 Mod 1 - LD 139380
Arming Cell Mark 2 Mod 2 - LD 139381

3.1.2 The requirements and procedures established in OP 400 shall be complied with as applicable.

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3.1.3 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample will be tested for the purpose of determining compliance with the requirements of this specification, the drawings, and the contract. See 4.1.2.

3.1.4 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Ordnance.

3.1.5 Conflicting requirements. - Conflicting requirements arising between this specification and any other specifications, publications, or drawings listed herein, shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.1.6 Soldering. - Unless otherwise specified, soldering processes shall conform to the requirements specified in Military Specification MIL-S-6872. Electrical connections shall be mechanically secure before soldering wherever possible. Only rosin flux, containing no acid, shall be used. The use of rosin-core solder is permitted.

3.2 Performance requirements and product characteristics. - The arming cell covered by this specification shall meet the following performance requirements and product characteristics.

3.2.1 Silver electrode. - The silver electrode shall consist of a pure silver cup, composed of at least 99 percent silver, filled with at least 5 grams of chemically pure fused silver chloride. The silver chloride shall not crack or chip nor become loose or fall out of the cup. The surface of the silver chloride shall be clean and free from wax, grease, paint or other foreign matter.

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3.2.2 Zinc electrode. - The surface of the zinc electrode shall be free from wax, grease, paint, or other foreign matter.

(a) Mechanical cleaning. - The zinc element of the arming cell shall be cleaned of phenolic mold splash (or oxides) as specified in 4.6.3 (a).

(b) Degreasing. - The zinc element of the arming cell shall be cleaned of all traces of grease as specified in 4.6.3 (b).

3.2.3 Wax. - The wax used to fill the resistor well and conductor conduit, and to pot the silver cup, shall be one having a melting point no lower than 180°F and a minimum tendancy to soften at temperatures below this point.

3.2.4 Dissolving time.

3.2.4.1 Mod 1 only. - The dissolving time of the Arming Cell Mark 2 Mod 1 shall be 4 ±1 days when tested as specified in 4.6.5.1.

3.2.4.2 Mod 2 only. - The dissolving time of the Arming Cell Mark 2 Mod 2 shall be 6 ±2 hours when tested as specified in 4.6.5.2.

3.2.5 Internal resistance, Mod 1 only. - The resistance between the zinc plate and the silver chloride cup shall be 200 ohms ±5 percent when tested as specified in 4.6.6.

3.2.6 Continuity, Mod 2 only. - The arming cell shall show a closed circuit when tested as specified in 4.6.7.

3.3 Environmental requirements.

3.3.1 Temperature. - Arming cells covered by this specification shall be subjected to temperature conditions for storage as specified in 4.7.1. Thereafter, the arming cell shall meet all performance requirements and product characteristics of this specification.

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3.3.2 Vibration. - Arming cells covered by this specification shall be mounted and vibrated as specified in 4.7.2, and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.3.3 Mechanical shock. - Arming cells covered by this specification shall be subjected to the mechanical shock test as specified in 4.7.3, and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.4 Workmanship and cleanliness. - The arming cells shall be free from dust, dirt, metal filings, moisture, solder splashes, chips, and other foreign matter. Necessary precautions shall be taken during manufacture and assembly to insure these conditions. The inclusion of foreign matter, and particularly the contamination of the electro-chemical plates with grease or moisture, will seriously affect the operation of the cell.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent specifications and drawings. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.1.1 The sampling plan required for the dissolving test as specified in paragraph 4.6.5, shall be in accordance with Inspection level L5, Table VII of the Appendix to MIL-STD-105, Sampling For Expensive Testing By Attributes.

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4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of 20 arming cells shall be manufactured using the methods and procedures proposed for full production. The Government inspector shall perform, witness, or participate in such inspections and tests on the samples as are required for full production. Upon conditional acceptance by the inspector, the preproduction sample shall be forwarded, together with a report of inspection, to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.2.1 Preproduction sample for a subsequent contract. - The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory materials as covered by this document.

4.1.3 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.1.4 Periodic production samples. - Unless otherwise specified in the contract, requisition or order, a random sample of 3 arming cells will be selected by the Government inspector from each succeeding lot or grand lot of 500 or less, for submission to a Bureau of Ordnance assigned activity to determine compliance with the requirements of the contract, specification, and drawings. An arming cell shall be considered satisfactory if it meets all the tests specified in 4.6 and 4.7 inclusive. The failure of an arming cell to comply with any of the environmental test requirements shall

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result in the cessation of Government inspection until the manufacturer has corrected all deficiencies in the production process. At the discretion of the procuring activity, such a failure may result in the rejection of the material represented. The manufacturer may resubmit rejected lots, subsequent to the completion of satisfactory corrective action as determined by the procuring activity.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of arming cells covered by this document shall be classified as follows:

(a) Acceptance tests.
(b) Preproduction tests.
(c) Periodic production tests.

4.3.1 Acceptance tests. - Acceptance tests for acceptance of the arming cell shall consist of tests to determine compliance with all requirements specified in this document except the environmental criteria.

4.3.2 Preproduction and periodic production tests. - Preproduction and periodic production tests are those performed on samples submitted in accordance with 4.1.2 and 4.1.4 to determine that the lot and production meet all the requirements of this specification. Acceptance shall be based on no defects in a sample. Acceptable preproduction and periodic production sample units, shall not be released for service use and shall be stamped or marked as follows: "Sample - Not for Service Use".

4.4 Test equipment.

4.4.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

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4.4.1.1 Equipment required for performing the electrical tests is as follows:

(a) Ohmmeter capable of measuring 0.5 ohm and 200 ohms within an accuracy of ±5 percent.

4.5 Test conditions. - Unless otherwise specified, tests shall be performed at 80° ±15°F and ambient relative humidity.

4.6 Acceptance testing. - Unless otherwise specified, the manufacturer shall devise procedures and methods satisfactory to the Bureau of Ordnance for conducting the acceptance tests specified herein. The right is reserved by the Bureau of Ordnance or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification, the drawings, or the contract.

4.6.1 Visual inspection. - The arming cell and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions, soldering, and workmanship are in accordance with this document and the drawings.

4.6.2 Silver electrode. - The silver electrode shall be inspected to determine compliance with the requirements of paragraph 3.2.1.

4.6.3 Zinc electrode. - The zinc electrode shall be inspected to see that it is free from all foreign matter.

(a) Mechanical cleaning. - To determine compliance with 3.2.2 (a), the zinc element of the arming cell shall be cleaned mechanically to remove phenolic mold splash (or oxides) from the surfaces or from the holes through the disk.

(b) Degreasing. - To determine compliance with 3.2.2 (b), the zinc element shall be cleaned of all traces of grease. The cleaning may be performed by using a commercial trichloroethylene bath or an electrolytic cleaning process. If the later process

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is to be used, the electrolyte shall consist of 2 ounces of sodium carbonate, 2 ounces of sodium bicarbonate, and water to make one gallon. The bath shall be operated at about 150°F. The zinc element shall be made cathodic and sufficient current shall be passed through the bath to produce vigorous evolution of hydrogen. A current densisty of 10 amperes per square foot and an immersion period of one-half to three minutes should remove all traces of grease. Following the electrolytic treatment, the zinc disk shall be rinsed in warm water and then shall be washed for five minutes in cold, circulating water.

4.6.4 Wax. - The inspector shall perform such tests on the selected wax as are necessary to determine compliance with 3.2.3.

4.6.5 Dissolving time. - For this test, a sample shall be selected at random and forwarded to an activity designated by the Bureau of Ordnance to determine compliance with the requirements of 3.2.4. The sample shall be selected as specified in paragraph 4.1.1.1.

4.6.5.1 Mod 1 only. - The dissolving time of the Arming Cell Mark 2 Mod 1 shall be 4 ±1 days, as specified in 3.2.4.1, when the cell is immersed in salt water of 3 percent salinity at 70° ±5°F, with an axial load of 26-1/2 pounds applied to the bushing and ring.

4.6.5.2 Mod 2 only. - The dissolving time of the Arming Cell Mark 2 Mod 2 shall be 6 ±2 hours, as specified in 3.2.4.2, when tested under the same conditions as the Mod 1 above.

NOTE: If any of the samples fail to meet this this test, the entire lot shall be rejected or repaired and then tested according to the same sampling schedule. If any of these samples fail to meet the test, the entire lot shall be rejected.

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4.6.6 Internal resistance test, Mod 1 only. - The resistance between the zinc plate and the silver chloride cup of the arming cell shall be checked with test item 4.4.1.1 (a) to determine compliance with the requirements of 3.2.5. The internal resistance shall be 200 ohms ±10 percent.

4.6.7 Continuity test, Mod 2 only. - To determine compliance with 3.2.6, the arming cell shall show a closed circuit when tested with test item 4.4.1.1 (a).

NOTE: The Arming Cell Mark 2 Mod 2 does not have an internal resistor. The zinc plate and silver cup are connected directly together to complete the circuit.

4.7 Environmental tests. - Mechanisms subjected to the environmental conditons given below shall show no damage and shall thereafter be tested to determine compliance with all other requirements of this specification.

4.7.1 Temperature storage test. - The arming cell shall first be stored at a temperature of -70°F for a period of 24 hours. At the end of this time, the cell shall be placed in an oven, preheated to 140°F, and stored for 24 hours at this temperature. At the end of this time, the cell shall be conditioned back to 70°F. The arming cells shall sustain no damage which would prevent their proper operation.

4.7.2 Vibration, rigidly mounted. - The arming cell shall be mounted rigidly to a vibration table by means of a mounting method simulating the actual mounting used in service. The vertical displacement component of the vibration shall be simple harmonic motion. The samples shall be mounted on the vibration table so that equal quantities of devices experience vibration perpendicular to the plane of mounting and vibration parallel to the plane of mounting. The mechanism shall be vibrated through a range of frequencies from 700 to 3000 cycles per minute in 100 cycles per minute increments. The duration of the vibration of each increment shall be one hour. The amplitude of the vibration shall be in accordance with the following schedule.

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FREQUENCY (cycles per minute)	AMPLITUDE (inches)
700 to 1500	0.030 ± 0.002
1600 to 2000	0.020 ± 0.002
2100 to 2500	0.013 ± 0.001
2600 to 3000	0.009 ± 0.001

4.7.3 Mechanical shock test. - The arming cell shall be subjected to a continuous two-phase shock applied in any direction with respect to the mechanism. The first (impact) phase shall consist of a change of velocity of 5.5 to 7.0 feet per second taking place in from 0.2 to 0.4 millisecond. During the second (drag) phase, the device shall be accelerated for a minimum time of 15 milliseconds in such a manner that during this time, the minimum average acceleration is 100 gravity units. The peak acceleration shall be between 125 and 150 gravity units.

4.8 Acceptance criteria. - Failure of any arming cell to meet any of the requirements of this specification shall result in rejection of the arming cell. If the number of failing arming cells exceeds the number permitted by the associated criteria, the lot represented will be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. - (Not applicable).

5.1.1.2 Preservative application. - (Not applicable).

5.1.1.3 Unit packaging. - A multiple of arming cell assemblies (maximum of 6) shall be unit packaged Method II-d (less preservative application) conforming to the requirements of Military Specification MIL-P-116. A standard commercial size metal tear-strip can, fabricated in accordance with the material requirements of Military Specification MIL-C-10464, shall be employed for the unit container.

5.1.1.3.1 Sufficient fillers, fabricated from non-hygroscopic material, shall be utilized within the unit container, to separate each arming cell and insure a tight package.

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5.1.2 Level B. - (Not applicable).

5.1.3 Level C.

5.1.3.1 Unit packaging. - Arming cells shall be packaged in a manner which will afford adequate protection from deterioration and physical damage during shipment from the contractor or supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior containers. - Twenty-four (24) unit containers shall be packed in an overseas type wood-cleated plywood box or in a Class 2, Style 2, 2 1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specifications PPP-B-601 or PPP-B-621, respectively.

5.2.1.2 A fiberboard liner, bottom pad and partitioned separators, fabricated from fiberboard material Class 2, W5c compliance symbol conforming to the material requirements of Federal Specification PPP-B-636, shall be incorporated within the shipping container to provide unit protection and prevent damage to contents. Top filler pads of the same material shall be employed, prior to box closure, as required to provide a tight pack.

5.2.1.3 When nailed wood boxes are used for the shipping container, Groups I, II or III woods shall be employed in box fabrication. Lumber shall be of one (1) inch nominal thickness and surfaced on both sides.

5.2.2 Level B. - (Not applicable).

5.2.3 Level C.

5.2.3.1 Arming cells shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers used shall conform to applicable carrier rules and regulations.

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5.3 Marking.

5.3.1 Normal marking. - In addition to the markings required by contract or order, unit packages, intermediate packages (when used), and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the title, number and date of this specification, and exceptions to this specification, applicable drawings and other documents. Procurement documents should also specify the desired level of packaging and packing. Level C should not be selected unless it is definitely known that the equipment will be used immediately upon receipt.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever: and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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